Coasters and pendulums

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use height differences to predict speed in ideal motion.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 9.*
ΔU=+12 J; gained. Less negative means greater energy.

02. Worked example: explain the model
U converts to K then back to U; equal ideal turning heights, zero speed at each turning point.

03. Read and interpret the model
Same final speed if no rotation or losses; times can differ.

04. Apply the physics
60 J lost U, 60 J gained K.

05. Explain your reasoning
h is vertical height difference, not distance along track; measure perpendicular to level reference.

06. Review: connect an earlier idea *Review: Unit 3, Day 14.*
24 m. The constant-Δv straight-line model omits full orbital dynamics.

07. Exit check
12 J.

08. STRETCH: test the model
Initial 8 J, final 6 J, loss 2 J to thermal/sound/etc.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 9.*
U_i=-35 J; U_f=-15 J; ΔU=+20 J; gravity work=-20 J.

02. Worked example: explain the model
U converts to K then back to U; equal ideal turning heights, zero speed at each turning point.

03. Read and interpret the model
Same final speed if no rotation or losses; times can differ.

04. Apply the physics
v^2=36+2(10)(5)=136; v 11.7 m/s.

05. Explain your reasoning
h is vertical height difference, not distance along track; measure perpendicular to level reference.

06. Review: connect an earlier idea *Review: Unit 3, Day 14.*
J=12 N·s; v_f≈0 m/s from $mv_i+J=mv_f$.

07. Exit check
No, not from rest and without input; would require negative K at 6 m.

08. STRETCH: test the model
Top height 4; v^2=80, v≈8.94. N/m=v^2/R-g=30 N/kg, positive contact.